

MAT301

Complete Lecture Notes

Matthew Farah

May 13, 2026

Contents

1	Introduction
----------	---------------------

2

Lecture 1

Introduction

- Notes about the course:
 - 4 problems (5% each).
 - Weekly quizzes (10% total).
 - 2 midterms (15% each).
 - Exam (40% total).
- Algebra is concerned with studying the implications that certain collections of axioms have on solving problems.

Definition 1: Let G be a set. A binary operator $*$ is a function $*$: $G \times G \rightarrow G$.

Definition 2: Let G be a set. G is a group under a binary operator $*$ if $(G, *)$ satisfies all 3 of the following criteria:

1. Associativity:
 - $\forall a, b, c \in G; (a * b) * c = a * (b * c)$.
2. Existence of an identity element:
 - $\exists e \in G; \forall a \in G; a * e = e * a = a$.
3. Existence of inverses:
 - $\forall a \in G; \exists b \in G; a * b = b * a = e$.

Definition 3: A group $(G, *)$ is called abelian if, in addition to satisfying the group axioms, $*$ satisfies the commutativity of $*$ on G :

$$\forall a, b \in G; a * b = b * a$$

Example 1: Which of the following are groups?

(a) $\mathbb{Z}$ under addition?

Solution:

Yes!

(b) $\mathbb{N}$ under addition?

Solution:

No! There is no inverse for any element except 0 if you let $0 \in \mathbb{N}$.

(c) $\mathbb{Z}$ under multiplication?

Solution:

No! Many elements, such as 2, do not have a multiplicative inverse in $\mathbb{Z}$.

(d) $\mathbb{Z}$ under subtraction?

Solution:

No! Subtraction isn't associative. For example:

$$(1 - 2) - 3 = -4 \neq 2 = 1 - (2 - 3)$$

Example 2: Do any of the following sets under addition constitute a group?

$$\{1, 2, 3\}, \quad \{0, 1, 2, 3\}, \quad \{-3, -2, -1, 0, 1, 2, 3\}$$

Solution:

No, since none of them are closed under addition. For example, $1 + 3 = 4$ does not belong to any of the sets.